AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

(AIUB)

FACULTY OF SCIENCE & TECHNOLOGY



Course Title:

INTRODUCTION TO DATABASE (2108)

Semester: Spring, 2023-2024

Section: [I]

Project Title:

Metro Rail Management System

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Introduction

The Metro Rail Management System is a cutting-edge system that seamlessly integrates essential capabilities to improve metropolitan transit. To accommodate

passengers, the system provides effective station administration, facilitating easy navigation and boarding procedures. Recharge and pass card features streamline fare transactions, and seat reservations guarantee comfort while traveling. The Lost and Found service helps quickly recover misplaced things by adding an extra degree of security. Passengers can also offer their ideas and experiences through the feedback mechanism, which promotes ongoing improvement. The goal of this all-inclusive system is to provide a quick, safe, and responsive metro transit experience.

Scenario Description

<table>
<tr><td colspan="2">StudentID1: 23-51384-1
Name: MD. ZARIF IQBAL
StudentID3: 23-51372-1
Name: MOMINUL ISLAM</td></tr>
<tr><td colspan="2">StudentID2: 23-51373-1
Name: D.M. SADMAN SAKIB
StudentID4: 23-51308-1
Name: PRITHILA RANI SAHA</td></tr>
<tr><td colspan="2">StudentID5: 23-51396-1
Name: MD. EKRAMUL ANJUM SABID

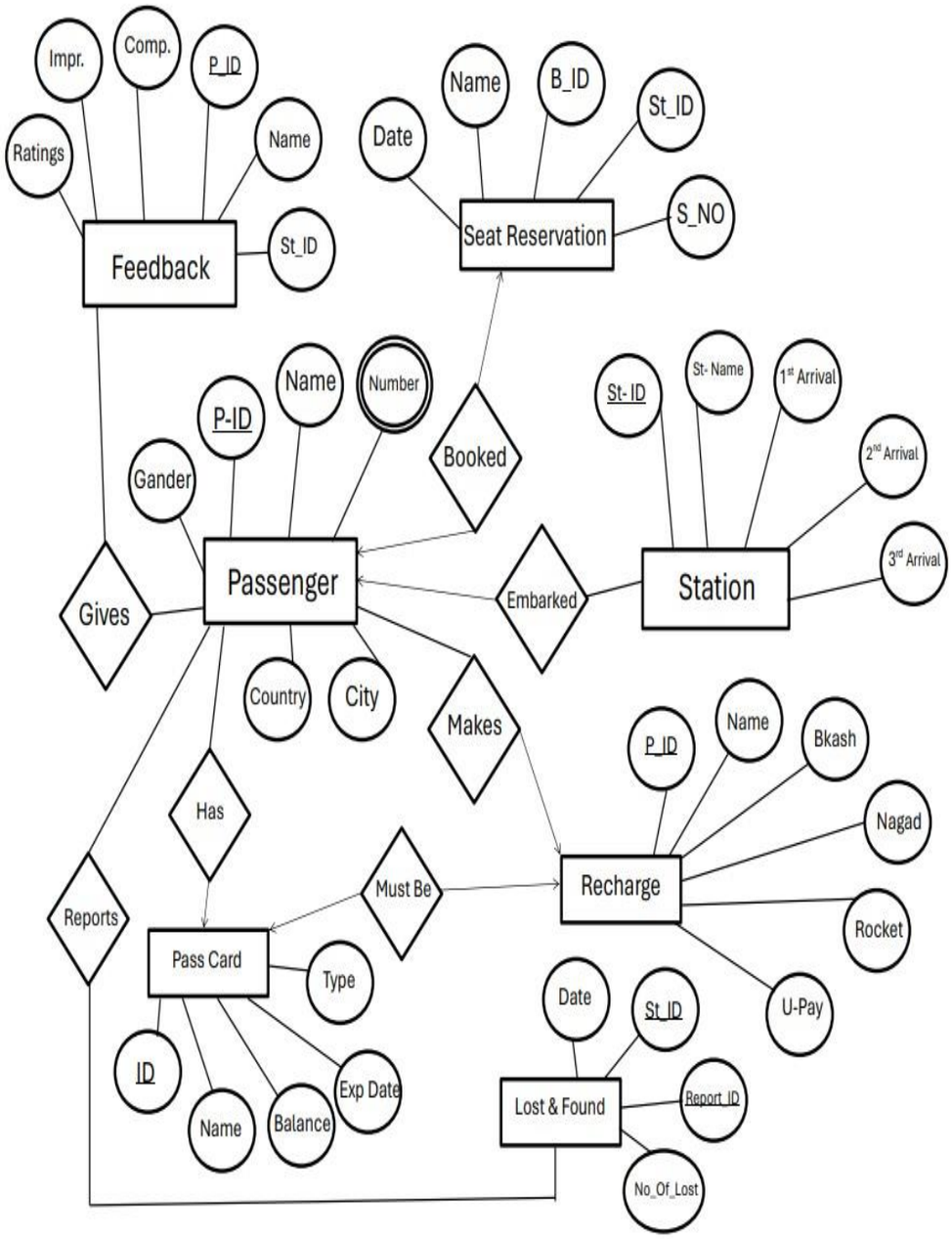
CO2: Understand the fundamental concepts underlying database systems and gain hands-on experience with ER diagram Case study</td></tr>
<tr><td>PO-c2: Develop process for complex computer science and engineering problems considering cultural and societal factors.</td><td>Marks</td></tr>
</table>

Scenario Description

In a Metro Rail Management System, A passenger embarked on their journey towards the station. Many passengers can visit a single station at a particular time. A passenger is identified by their Passenger id. The system also stores their name, number and gender. A Station is identified by its station id and station name. It has also shown metro's departure time. Pass-cards are issued to passengers for fare payment and access. A passenger may have many pass-cards. A passenger can have at most one payment method. Pass-Card carries Passenger id, expiry date, balance, type and a pass-card is assigned to an individual user. Pass-cards must be recharged by a passenger. But passenger can recharge pass-card by many ways Bkash, Nagad, Rocket, U-pay. It also carries passengers' IDs and names. Seat reservation enables passengers to book seats in advance. Reserved seats can be identified by their booked id, passenger id, seat number, station name and reservation date. And one passenger can have multiple bookings also multiple booking can be belong to one passenger. Multiple bookings can be made for the same station. And one station can have multiple bookings. If any passengers lost their stuff in the metro, they could report

it to the Lost & found counter. Lost report is identified by their report id, report date and number of lost items. A passenger can have at most one lost and found report. Following their metro journey, passengers can share their thoughts by providing feedback. Feedback permits travelers to offer complaints, suggestions for enhancements and evaluations. The Metro rail authority has the capability to identify individuals providing feedback, recording both passenger Id / identification and names. Multiple feedback can be belonged to one passenger.

ER Diagram



Normalization Embarked:

UNF:

1. P_ID, Name, Number, Gender, City,Country, St_ID, St_Name, Dep_Time

1NF:

Number is Multi Valued Attribute.

1. P_ID, Name, Number, Gender, City,Country, St_ID, St_Name, Dep_Time

2NF:

1. P_ID, Name, Number, Gender, City,Country
2. St_ID, St_Name, Dep_Time

3NF:

1. P_ID, Name, Number, Gender
2. St_ID, St_Name, Dep_Time
3. A_Id City, Country

Table Creation:

1. P_ID, Name, Number, Gender
2. St_ID, St_Name, Dep_Time , P_Id (FK)
3. A_Id, City, Country

Makes:

UNF:

1. P_ID, Name, Number, Gender, City,Country ,Bkash , Nagad , Rocket , Upay

1NF:

Number is Multi Valued Attribute.

1. P_ID, Name, Number, Gender, City,Country, Bkash , Nagad , Rocket , Upay

2NF:

1. P_ID, Name, Number, Gender, City,Country
2. Bkash , Nagad , Rocket , Upay

3NF:

1. P_ID, Name, Number, Gender
2. St_ID, St_Name, Dep_Time
3. A_Id, City,Country

Table Creation:

1. P_ID, Name, Number, Gender
2. Bkash , Nagad , Rocket , Upay , A_ID , P_ID(FK)
3. A_Id, City ,Country

Booked:

UNF:

1. P_ID, Name, Number, Gender, City,Country, B_Id , Seat_No, Date

1NF :

There is no Multi Valued Attribute

1. P_ID, Name, Number, Gender, City,Country, B_Id , Seat_No, Date

2NF :

1. P_ID, Name, Number, Gender, City,Country

2 . B_Id , Seat_No, Date

3NF :

1. P_ID, Name, Number, Gender
2. B_Id, Seat_No, Date
3. A_Id, City,Country

Table Creation:

1. P_ID, Name, Number, Gender
2. B_Id, Seat_No, Date , P_Id (Fk)
3. A_Id, City,Country

Has:

UNF:

1. P_ID, Name, Number, Gender, City, Country ,Pass_ID , Balance , Exp_Date , Type

1NF:

There is no Multi Valued Attribute.

1. P_ID, Name, Number, Gender, City,Country, Pass_ID , Balance , Exp_Date , Type

2NF:

1. P_ID, Name, Number, Gender, City,Country
2. Pass_ID , Balance , Exp_Date , Type

3NF:

1. P_ID, Name, Number, Gender
2. Pass_ID , Balance , Exp_Date , Type

3. A_Id , City,Country

Table Creation:

1. P_ID, Name, Number, Gender
2. Pass_ID , Balance , Exp_Date , Type , P_Id (FK)
3. A_Id, City, Country

Must be :

UNF:

1. Pass_ID, Name, Balance,Exp_Date,Type, Recharge_Id , Bkash, Nagad, Rocket, Upay

1NF :

There is no Multi Valued Attribute

1. Pass_ID(PK), Name, Balance,Exp_Date,Type, Recharge_Id(PK) , Bkash, Nagad, Rocket, Upay

2NF :

1. Pass_ID(PK), Name, Balance,Exp_Date,Type

2 . Recharge_Id (PK) , Bkash, Nagad, Rocket, Upay

3NF :

1. Pass_ID (PK) , Name, Balance,Exp_Date,Type

2 . Recharge_Id (PK) , Bkash, Nagad, Rocket, Upay, Pass_Id (FK)

Table Creation :

1.Pass ID , Name, Balance,Exp_Date,Type

2 . Recharge_Id (PK) , Bkash, Nagad, Rocket, Upay, Pass_Id (FK)

Gives :

UNF:

1. P_ID, Name, Number, Gender, City,Country, F_Id , Name , Complain , Improvement , Rating

1NF :

There is no Multi Valued Attribute

1. P_ID, Name, Number, Gender, City,Country, F_Id , Name , Complain , Improvement , Rating

2NF :

1. P_ID, Name, Number, Gender, City,Country
2 . F_Id , Complain , Improvement , Rating

3NF :

1. P_ID, Name, Number, Gender
2. F_Id , Complain , Improvement , Rating , P_Id (FK)
3. A_Id, City,Country

Table Creation:

1. P_ID, Name, Number, Gender
2. F_Id , Complain , Improvement , Rating , P_Id (FK)
3. A_Id, City,Country

Reports:

UNF:

1. P_ID, Name, Number, Gender, City,Country, R_Id , Date , No_of_lost_item

1NF :

There is no Multi Valued Attribute

1. P_ID, Name, Number, Gender, City,Country, R_Id , Date , No_of_lost_item

2NF :

1. P_ID, Name, Number, Gender, City,Country
2 . R_Id , Date , No_of_lost_item

3NF :

1. P_ID, Name, Number, Gender
2. R_Id , Date , No_of_lost_item, P_Id (FK)
3. A_Id, City,Country

Table Creation:

1. P_ID, Name, Number, Gender
2. R_Id , Date , No_of_lost_item, P_Id (FK)
3. A_Id, City,Country

Finalization

1. P_Id (Pk) , Name , Number , Gender
2. St_Id (Pk) , St_Name , Departure_Time , **P_Id (Fk)**
3. A_Id (Pk) , City , Country , **P_ID(FK)**
4. Recharge_Id (Pk) , Bkash , Nagad, Rocket , Upay , **P_Id (Fk)**
5. B_Id , Seat_No , Date , **P_Id (Fk)**
6. Pass_Id , Balance , Exp_Date ,Type , **P_Id (Fk)**
7. F_Id (Pk) , Complains , Improvement , Rating , **P_id (Fk)**
8. Report_Id , Date , No_of_lost_item , **P_Id (FK)**

Schema Diagram

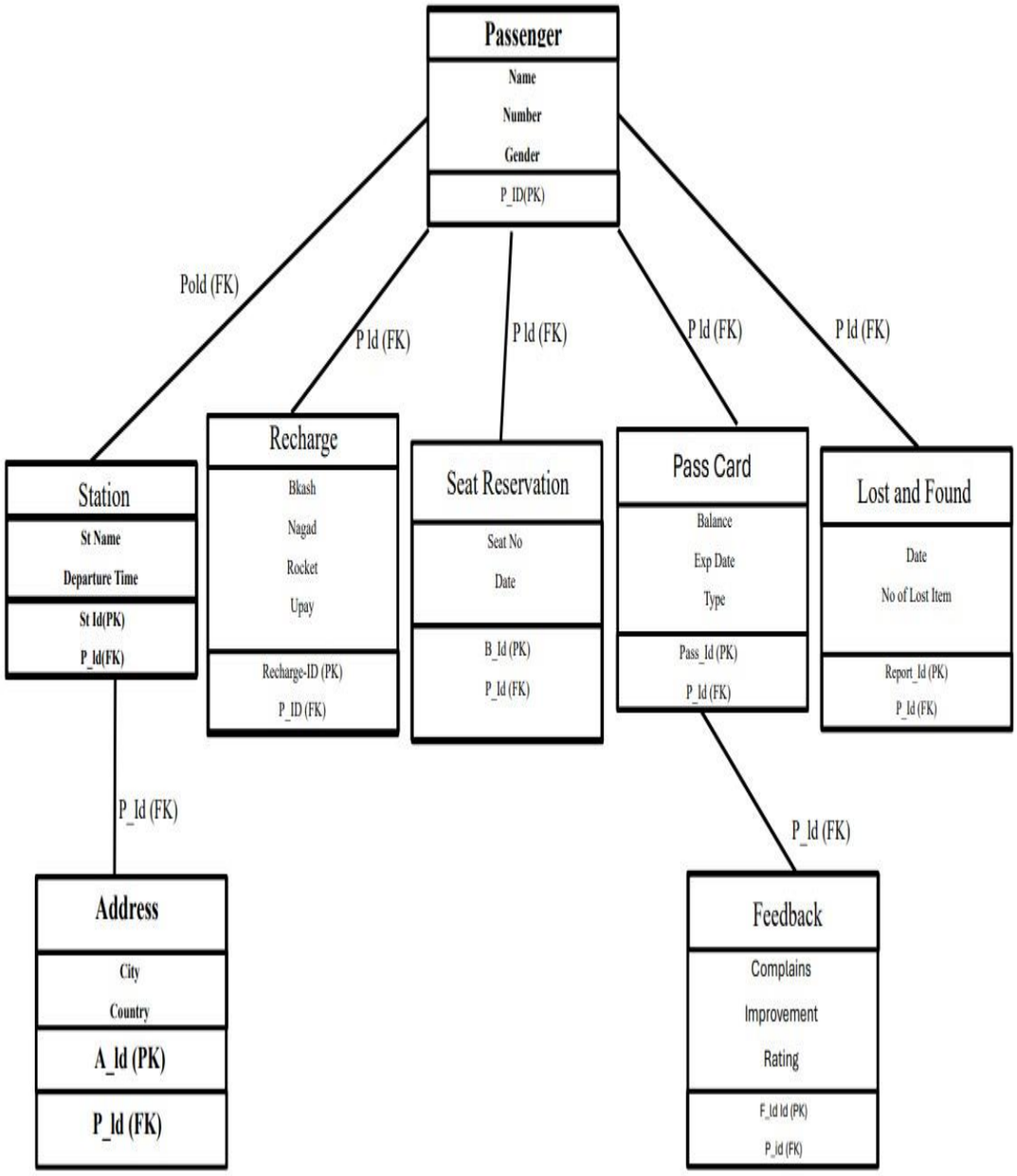


Table Creation

StudentID1: 23-51384-1 Name: MD. ZARIF IQBAL	StudentID3: 23-51372-1 Name: MOMINUL ISLAM
StudentID2: 23-51373-1 D.M. SADMAN SAKIB	StudentID4: 23-51308-1 Name: Name: PRITHILA RANI SAHA
StudentID5: 23-51396-1 Name: MD. EKRAMUL ANJUM SABID	
CO4: Creating DML, DDL using Oracle and connection with ODBC/JDBC for existing JAVA application	
PO-e-2: Use modern engineering and IT tools for prediction and modeling of computer science and engineering problem	Marks complex

1. PASSENGER

```
CREATE TABLE PASSENGER (
    P_ID NUMBER(5) CONSTRAINT PK_PASSENGER PRIMARY KEY,
```

P_NAME VARCHAR2(15),

PhoneNumber VARCHAR2(15),

Gender VARCHAR2(10)

);

```
desc PASSENGER
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **PASSENGER**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PASSENGER	P_ID	Number	-	5	0	1	-	-	-
	P_NAME	Varchar2	15	-	-	-	✓	-	-
	PHONENUMBER	Varchar2	15	-	-	-	✓	-	-
	GENDER	Varchar2	10	-	-	-	✓	-	-

1 - 4

2. STATION

CREATE TABLE STATION (

Station_ID NUMBER(5) CONSTRAINT PK_STATION PRIMARY KEY,

P_ID NUMBER(5),

Station_NAME VARCHAR2(15),

Departure_Time NUMBER(15),

Gender VARCHAR2(10),

CONSTRAINT FK_PID FOREIGN KEY (P_ID) REFERENCES PASSENGER(P_ID));

```
desc STATION
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **STATION**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
STATION	STATION_ID	Number	-	5	0	1	-	-	-
	P_ID	Number	-	5	0	-	✓	-	-
	STATION_NAME	Varchar2	15	-	-	-	✓	-	-
	DEPARTURE_TIME	Number	-	15	0	-	✓	-	-
	GENDER	Varchar2	10	-	-	-	✓	-	-

1 - 5

3. RECHARGE

CREATE TABLE RECHARGE (

Recharge_ID NUMBER(5) CONSTRAINT PK_RECHARGE PRIMARY KEY,

P_ID NUMBER(5),

BKASH VARCHAR2(15),

NOGOD VARCHAR2(15),

ROCKET VARCHAR2(10),

UPAY VARCHAR2(10),

CONSTRAINT FK_RECHARGE_PID FOREIGN KEY (P_ID) REFERENCES PASSENGER(P_ID)

);

```
desc RECHARGE
```

Results Explain Describe Saved SQL History

Object Type TABLE Object RECHARGE

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
RECHARGE	RECHARGE_ID	Number	-	5	0	1	-	-	-
	P_ID	Number	-	5	0	-	✓	-	-
	BKASH	Varchar2	15	-	-	-	✓	-	-
	NOGOD	Varchar2	15	-	-	-	✓	-	-
	ROCKET	Varchar2	10	-	-	-	✓	-	-
	UPAY	Varchar2	10	-	-	-	✓	-	-

1 - 6

4. RESERVATION

CREATE TABLE RESERVATION (

Booked_ID NUMBER(5) CONSTRAINT PK_RESERVATION PRIMARY KEY,

P_ID NUMBER(5),

Seat_No VARCHAR2(15),

Reservation_Date DATE,

CONSTRAINT FK_RESERVATION_PID FOREIGN KEY (P_ID) REFERENCES PASSENGER(P_ID));

```
desc RESERVATION
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **RESERVATION**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
RESERVATION	BOOKED_ID	Number	-	5	0	1	-	-	-
	P_ID	Number	-	5	0	-	✓	-	-
	SEAT_NO	Varchar2	15	-	-	-	✓	-	-
	RESERVATION_DATE	Date	7	-	-	-	✓	-	-

1 - 4

5. PASSCARD

CREATE TABLE PASSCARD (

Pass_ID NUMBER(5) CONSTRAINT PK_PASSCARD PRIMARY KEY,

P_ID NUMBER(5),

Balance NUMBER(15),

Expire_Date DATE,

Type VARCHAR2(10),

CONSTRAINT FK_PASSCARD_PID FOREIGN KEY (P_ID) REFERENCES PASSENGER(P_ID));

```
desc PASSCARD
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **PASSCARD**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PASSCARD	PASS_ID	Number	-	5	0	1	-	-	-
	P_ID	Number	-	5	0	-	✓	-	-
	BALANCE	Number	-	15	0	-	✓	-	-
	EXPIRE_DATE	Date	7	-	-	-	✓	-	-
	TYPE	Varchar2	10	-	-	-	✓	-	-

1 - 5

6. LOST_AND_FOUND

```
CREATE TABLE LOST_AND_FOUND (
    Report_ID NUMBER(5) CONSTRAINT PK_LOST_AND_FOUND PRIMARY KEY,
    P_ID NUMBER(5),
    Number_of_Lost_Items NUMBER(15),
    Report_Date DATE,
    CONSTRAINT FK_LOST_AND_FOUND_PID FOREIGN KEY (P_ID) REFERENCES
PASSENGER(P_ID)
);
```

```
desc LOST_AND_FOUND
```

Results Explain **Describe** Saved SQL History

Object Type **TABLE** Object **LOST_AND_FOUND**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
LOST_AND_FOUND	REPORT_ID	Number	-	5	0	1	-	-	-
	P_ID	Number	-	5	0	-	✓	-	-
	NUMBER_OF_LOST_ITEMS	Number	-	15	0	-	✓	-	-
	REPORT_DATE	Date	7	-	-	-	✓	-	-

1 - 4

7. FEEDBACK

```
CREATE TABLE FEEDBACK (
F_ID NUMBER(5) CONSTRAINT PK_FEEDBACK PRIMARY KEY,
P_ID NUMBER(5),
Complains VARCHAR2(50),
Improvements VARCHAR2(50),
Rating NUMBER(10),
CONSTRAINT FK_FEEDBACK_PID FOREIGN KEY (P_ID) REFERENCES PASSENGER(P_ID));
```

```
desc FEEDBACK
```

Results | Explain | Describe | Saved SQL | History

Object Type **TABLE** Object **FEEDBACK**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
FEEDBACK	F_ID	Number	-	5	0	1	-	-	-
	P_ID	Number	-	5	0	-	✓	-	-
	COMPLAINS	Varchar2	50	-	-	-	✓	-	-
	IMPROVEMENTS	Varchar2	50	-	-	-	✓	-	-
	RATING	Number	-	10	0	-	✓	-	-

1 - 5

8. ADDRESS

```
CREATE TABLE ADDRESS (
    A_ID NUMBER(5) CONSTRAINT PK_ADDRESS PRIMARY KEY,
    P_ID NUMBER(5),
    Country VARCHAR2(50),
    City VARCHAR2(50),
    CONSTRAINT FK_ADDRESS_PID FOREIGN KEY (P_ID) REFERENCES PASSENGER(P_ID)
);
```

```
desc ADDRESS
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **ADDRESS**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
ADDRESS	A_ID	Number	-	5	0	1	-	-	-
	P_ID	Number	-	5	0	-	✓	-	-
	COUNTRY	Varchar2	50	-	-	-	✓	-	-
	CITY	Varchar2	50	-	-	-	✓	-	-

1 - 4

Insert Values in Table

DATA INSERTATION:

1. PASSENGER

```
INSERT INTO PASSENGER VALUES
(1,'ZARIF',01999275845,'MALE');
INSERT INTO PASSENGER VALUES
(2,'SADMAN',01318087759,'MALE');
INSERT INTO PASSENGER VALUES
(3,'SABID',01717973101,'MALE');
INSERT INTO PASSENGER VALUES
(4,'PRITHILA',01314401264,'FEMALE');
INSERT INTO PASSENGER VALUES
(5,'MOEIN',01746033170,'MALE');
```

P_ID	P_NAME	PHONENUMBER	GENDER
1	ZARIF	1999275845	MALE
2	SADMAN	1318087759	MALE
3	SABID	1717973101	MALE
4	PRITHILA	1314401264	FEMALE
5	MOEIN	1746033170	MALE

2. STATION

```
INSERT INTO STATION VALUES
(101, 1, 'DIABARI',8.00);
INSERT INTO STATION VALUES
(102, 2, 'MIRPUR',9.00);
INSERT INTO STATION VALUES
(103, 3, 'AGARGAON',10.00);
INSERT INTO STATION VALUES
(104, 4, 'FARMGATE',11.00);
INSERT INTO STATION VALUES
(105, 5, 'DHAKAUNIVERSITY',12.00);
```

STATION_ID	P_ID	STATION_NAME	DEPARTURE_TIME
101	1	DIABARI	8
102	2	MIRPUR	9
103	3	AGARGAON	10
104	4	FARMGATE	11
105	5	DHAKAUNIVERSITY	12

3. RECHARGE

```
INSERT INTO RECHARGE VALUES
(11, 1, 'YES',NULL,NULL,NULL);
INSERT INTO RECHARGE VALUES
(22, 2, NULL,'YES',NULL,NULL);
INSERT INTO RECHARGE VALUES
(33, 3, NULL,NULL,'YES',NULL);
INSERT INTO RECHARGE VALUES
(44, 4, NULL,NULL,NULL,'YES');
INSERT INTO RECHARGE VALUES
(55, 5, 'YES',NULL,NULL,NULL);
```

RECHARGE_ID	P_ID	BKASH	NOGOD	ROCKET	UPAY
11	1	YES	-	-	-
22	2	-	YES	-	-
33	3	-	-	YES	-
44	4	-	-	-	YES
55	5	YES	-	-	-

5 rows returned in 0.00 seconds CSV Export

4. RESERVATION

```
INSERT INTO RESERVATION VALUES
(111, 1, '1A', '14-MAY-24');
INSERT INTO RESERVATION VALUES
(222, 2, '2B', '13-APRIL-24');
INSERT INTO RESERVATION VALUES
(333, 3, '3C', '12-MARCH-24');
INSERT INTO RESERVATION VALUES
(444, 4, '4D', '17-MAY-24');
INSERT INTO RESERVATION VALUES
(555, 5, '5E', '20-JUNE-24');
```

BOOKED_ID	P_ID	SEAT_NO	RESERVATION_DATE
222	2	2B	13-APR-24
333	3	3C	12-MAR-24
444	4	4D	17-MAY-24
555	5	5E	20-JUN-24
111	1	1A	14-MAY-24

5. PASSCARD

```
INSERT INTO PASSCARD VALUES
(10, 1, 999,'14-MAY-30','VIP');
INSERT INTO PASSCARD VALUES
(20, 2, 100,'14-MAY-27','REGULAR');
INSERT INTO PASSCARD VALUES
(30, 3, 99,'20-DEC-27','REGULAR');
INSERT INTO PASSCARD VALUES
(40, 4, 333,'30-MAY-27','VIP');
INSERT INTO PASSCARD VALUES
(50, 5, 100,'09-MARCH-25','REGULAR');
```

PASS_ID	P_ID	BALANCE	EXPIRE_DATE	TYPE
10	1	999	14-MAY-30	VIP
20	2	100	14-MAY-27	REGULAR
30	3	99	20-DEC-27	REGULAR
40	4	333	30-MAY-27	VIP
50	5	100	09-MAR-25	REGULAR

6. LOST_AND_FOUND

```
INSERT INTO LOST_AND_FOUND VALUES
(21,1,5,'21-APRIL-24');
INSERT INTO LOST_AND_FOUND VALUES
(45,2,0,NULL);
INSERT INTO LOST_AND_FOUND VALUES
(13,3,0,NULL);
INSERT INTO LOST_AND_FOUND VALUES
(113,4,3,'31-MARCH-24');
INSERT INTO LOST_AND_FOUND VALUES
(131,5,0,NULL);
```

REPORT_ID	P_ID	NUMBER_OF_LOST_ITEMS	REPORT_DATE
113	4	3	31-MAR-24
131	5	0	-
21	1	5	21-APR-24
45	2	0	-
13	3	0	-

5 rows returned in 0.00 seconds CSV Export

7. FEEDBACK

```
INSERT INTO FEEDBACK VALUES
(84,1,NULL,NULL,5);
INSERT INTO FEEDBACK VALUES
(73,2,'TOO SLOW',NULL,5);
INSERT INTO FEEDBACK VALUES
(92,3,NULL,NULL,3);
INSERT INTO FEEDBACK VALUES
(13,4,NULL,'SHOULD MAINTAIN HYGIENE',4);
INSERT INTO FEEDBACK VALUES
(72,5,NULL,'I AM IMPRESSED',5);
```

F_ID	P_ID	COMPLAINS	IMPROVEMENTS	RATING
92	3	-	-	3
73	2	TOO SLOW	-	5
84	1	-	-	5
13	4	-	SHOULD MAINTAIN HYGIENE	4
72	5	-	I AM IMPRESSED	5

5 rows returned in 0.00 seconds CSV Export

8. ADDRESS

```
INSERT INTO ADDRESS VALUES
(1720, 1,'BANGLADESH','GAZIPUR');
INSERT INTO ADDRESS VALUES
(6570, 2,'BANGLADESH','RAJSHAHI');
INSERT INTO ADDRESS VALUES
(2020, 3,'BANGLADESH','ISLAMPUR');
INSERT INTO ADDRESS VALUES
(1703, 4,'BANGLADESH','NARSHINGDI');
INSERT INTO ADDRESS VALUES
(1206, 5,'BANGLADESH','DHAKA');
```

A_ID	P_ID	COUNTRY	CITY
1720	1	BANGLADESH	GAZIPUR
6570	2	BANGLADESH	RAJSHAHI
2020	3	BANGLADESH	ISLAMPUR
1703	4	BANGLADESH	NARSHINGDI
1206	5	BANGLADESH	DHAKA

1. Single Row

A) Display the Passenger name in Upper Case for Passenger SADMAN.

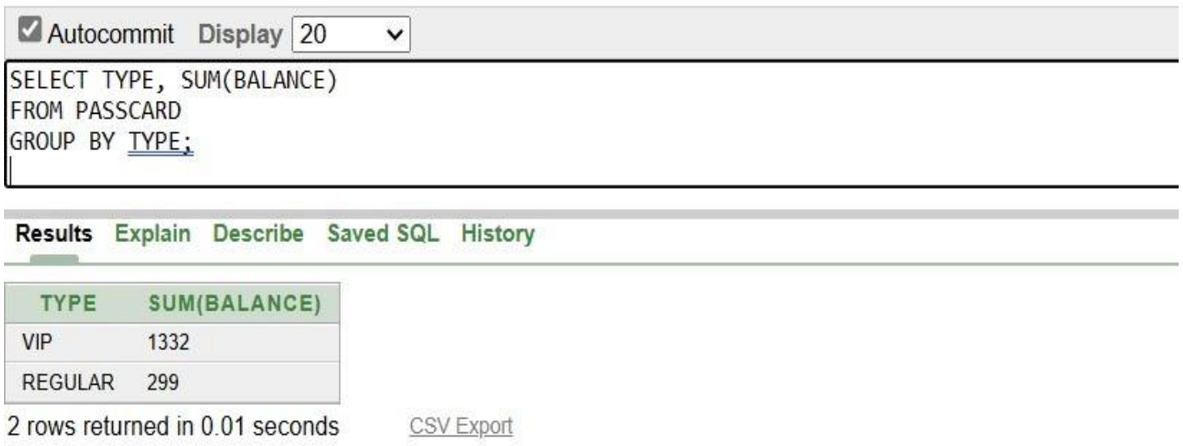


B) Display the Passenger name and number in one column and the new column name 'NAME_PHONE'.

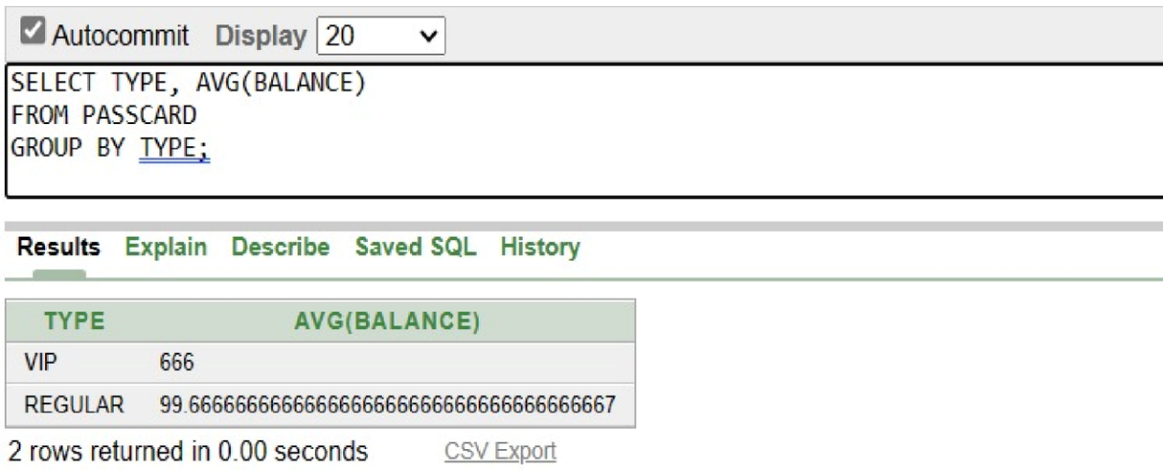


2.Group Function

A) Write a SQL query to display the total BALANCE for each type.

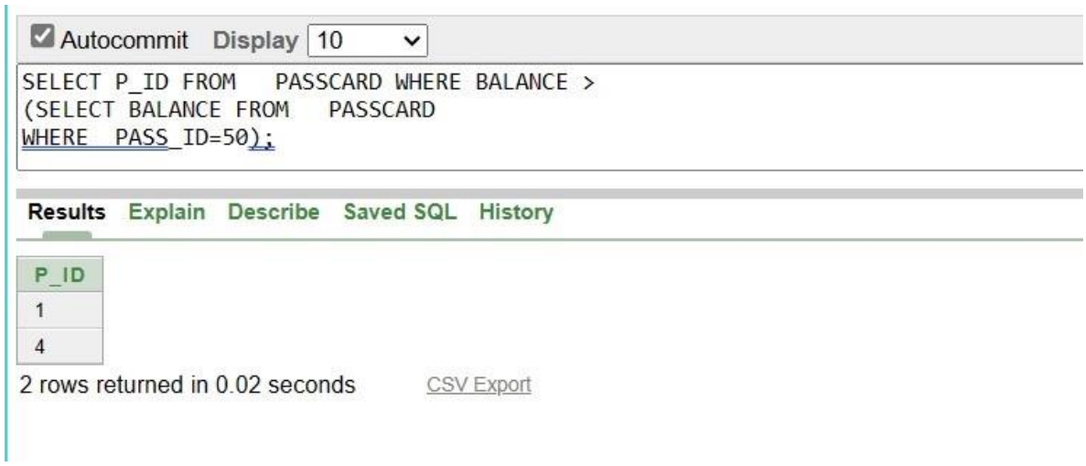


B) Write a SQL query to find the average balance for each type.

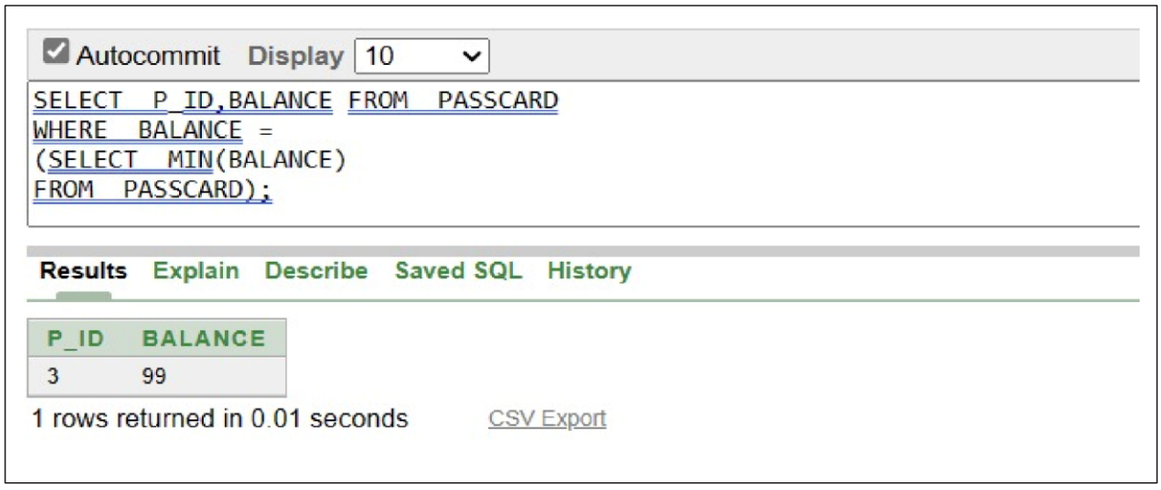


3.SUB QUERY

A) Display the names of passenger from who have higher Balance than the Passenger with a PASSCARD ID 50 .

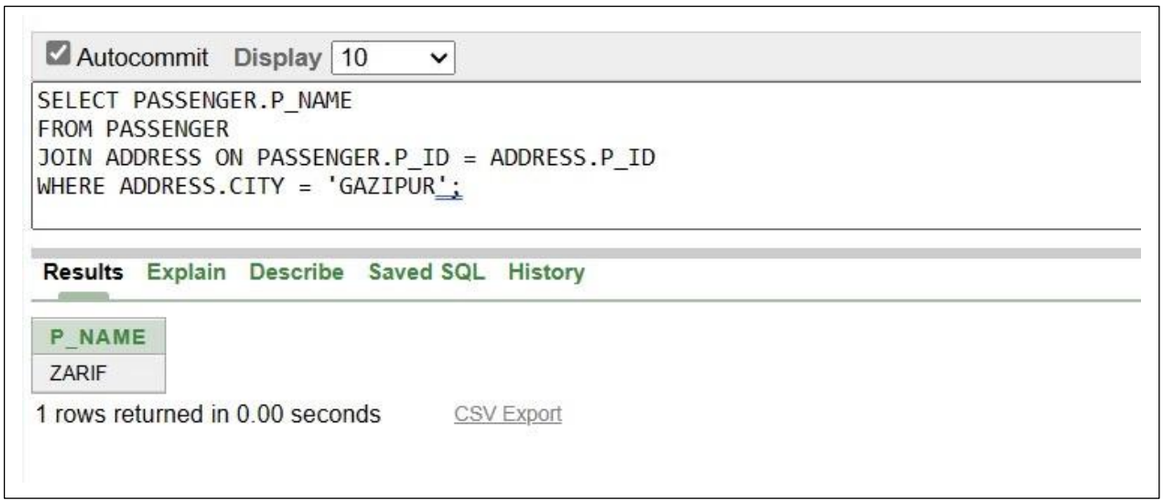


B DISPLAY the Passenger ID and Balance of Passenger who have the lowest Balance among all Passengers.

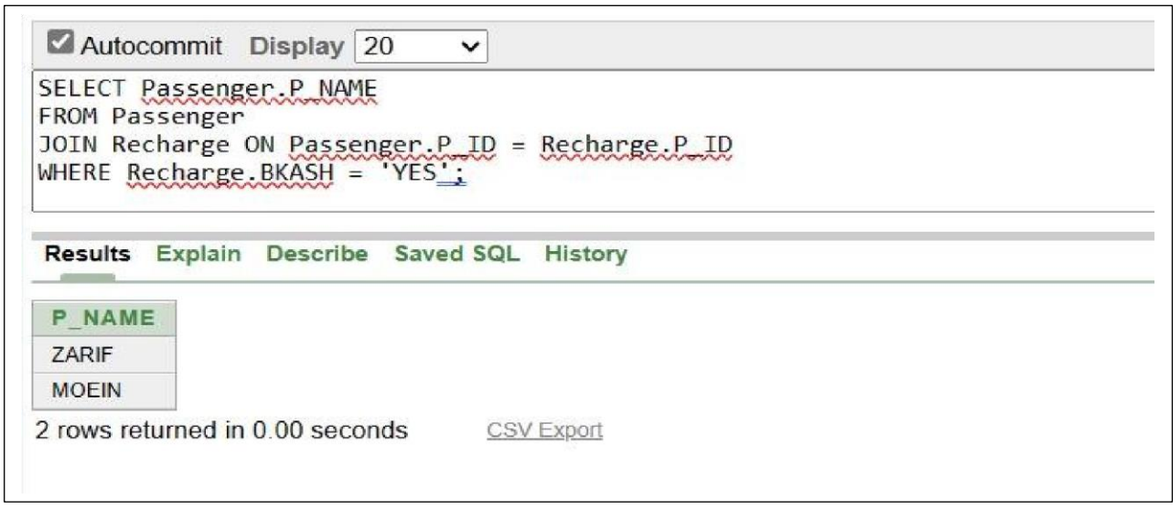


4. JOINING

A) Display the name of the PASSENGER who lived in Gazipur.



B) Display the name of the PASSENGER who recharged in BKASH.



5. VIEW

A) Create a view called PASSENGERView based on the NAME AND PHONE NUMBER from the PASSENGER table and Display the view.

Autocommit Display 10

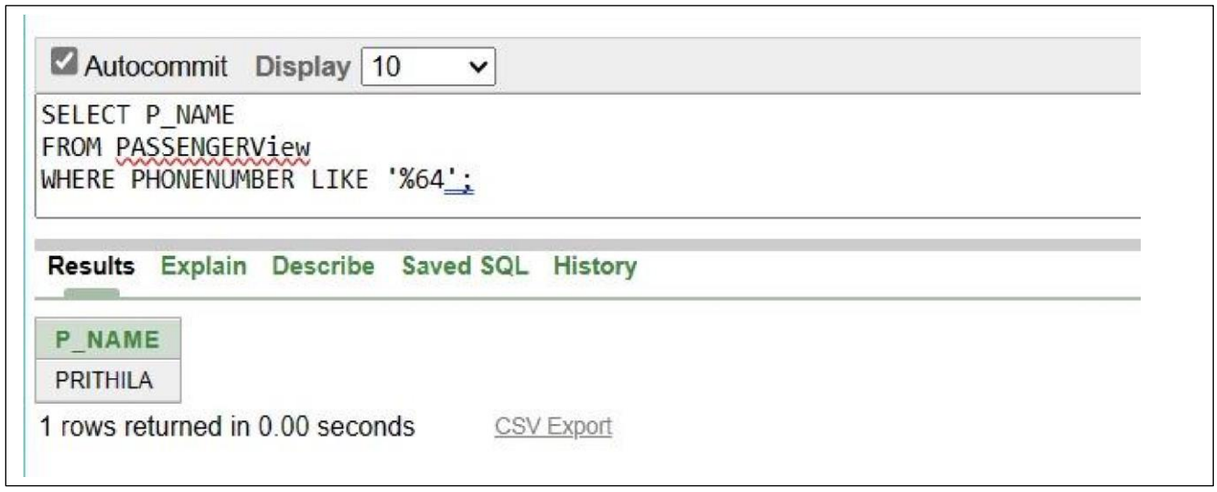
```
CREATE VIEW PASSENGERView AS SELECT P_NAME,PHONENUMBER FROM PASSENGER;
SELECT *FROM PASSENGERView;
```

Results Explain Describe Saved SQL History

P_NAME	PHONENUMBER
ZARIF	1999275845
SADMAN	1318087759
SABID	1717973101
PRITHILA	1314401264
MOEIN	1746033170

5 rows returned in 0.01 seconds CSV Export

B) Display the NAME of the passenger from the PASSENGERView view where the PHONE Number ends with 64 .

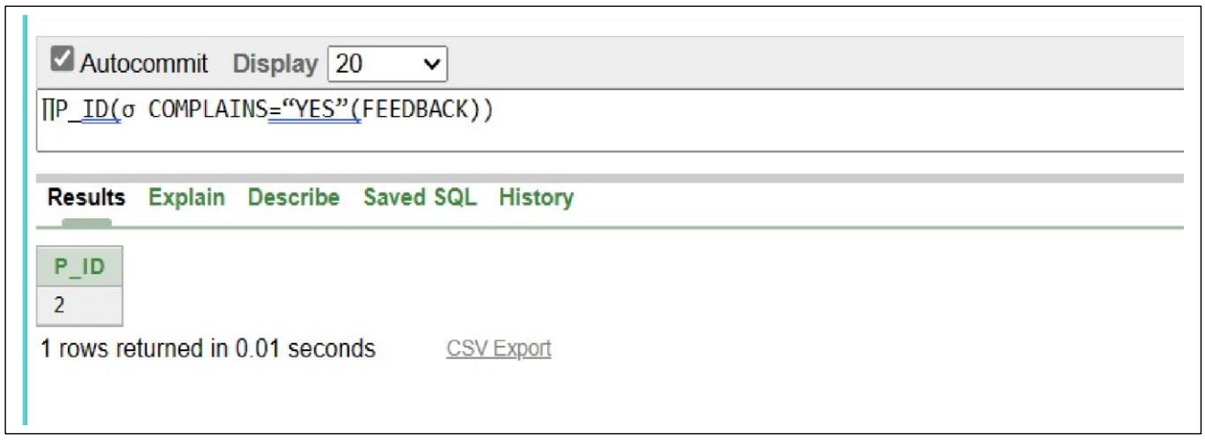


Relational Algebra

1. Display the passenger who lives in RAJSHAHI.



2. DISPLAY the passenger id who have complain about Metro Rail.



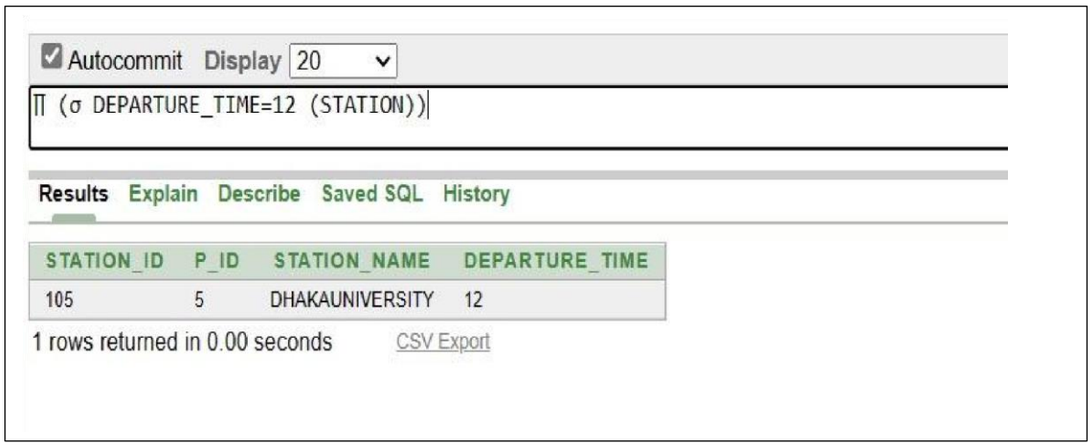
3. Display the passenger id and reservation date who have booked seat no 2B.



4. Display the passenger id who have more Balance than 100.



5. Display all the information about the Station which Departure Time is 12 pm .



Conclusion

In short, the Metro rail management system provides an integrated and efficient approach to urban transportation, improving the overall passenger experience with its comprehensive functionality.

The system effectively manages passenger flow, ensuring smooth access to stations and making travel easier.

With the top-up feature and registration card, passengers can easily manage their travel expenses, increase convenience and reduce congestion at ticket counters.

The seat reservation option further enhances comfort, allowing passengers to reserve their preferred seat in advance.

The Lost and Found feature ensures that any lost items can be tracked and returned efficiently, adding an extra layer of security and reliability to the system.

Additionally, a feedback mechanism allows passengers to express their concerns and suggestions, thereby promoting continuous improvement and meeting user needs.

Overall, the Metro Rail Management System presents itself as a powerful solution that not only streamlines operations but also prioritizes passenger satisfaction and convenience, demonstrating a modern approach and User-friendly for public transportation.

This comprehensive management system is ready to meet the dynamic demands of urban transportation, ensuring an enjoyable and reliable travel experience for all passengers.